

Haskell-Projects

I worked on Haskell to perform some basic functions to learn about functional programming.

Expected output for Quicksort using recursion

```
[1 of 1] Compiling Main                ( main.hs, main.o )
Linking a.out ...
Enter a list of numbers separated by spaces:
1 2 3 4 5 6
[1,2,3,4,5,6]
Enter a list of numbers separated by spaces:
6 5 4 3 2 1
[1,2,3,4,5,6]
Enter a list of numbers separated by spaces:
1 6 2 9 4 0 3
[0,1,2,3,4,6,9]
Enter a list of numbers separated by spaces:
-1 -6 9 -3 -12 54 -54
[-54,-12,-6,-3,-1,9,54]
Enter a list of numbers separated by spaces:
4 4 4 4 4 4 4
[4,4,4,4,4,4,4]

...Program finished with exit code 0
Press ENTER to exit console.
```

Expected output for Pattern

```
[1 of 1] Compiling Main                ( main.hs, main.o )
Linking a.out ...
  *
 ***
*****
 ***
  *

...Program finished with exit code 0
Press ENTER to exit console.
```

Expected output for Permutation and Combination Calculator

```

[1 of 1] Compiling Main                ( main.hs, main.o )
Linking a.out ...
Enter a number that will be n of nPk:
1
Enter a number that will be k of nPk:
2
Error: k must be less than or equal to n
Enter a number that will be n of nPk:
2
Enter a number that will be k of nPk:
1
The nPk is 2.0
The nCk is 2.0
Enter a number that will be n of nPk:
2
Enter a number that will be k of nPk:
2
The nPk is 2.0
The nCk is 1.0
Enter a number that will be n of nPk:
0
Enter a number that will be k of nPk:
0
The nPk is 1.0
The nCk is 1.0
Enter a number that will be n of nPk:
1
Enter a number that will be k of nPk:
0
The nPk is 1.0
The nCk is 1.0

```

Expected output for Sum of squares calculator

```

[1 of 1] Compiling Main                ( main.hs, main.o )
Linking a.out ...
Enter a number:
2
Enter a number:
4
The answer is 36
Enter a number:
1
Enter a number:
1
The answer is 4
Enter a number:
0
Enter a number:
0
The answer is 0
Enter a number:
-4
Enter a number:
-6
The answer is 100
Enter a number:
-1
Enter a number:
5
The answer is 16

...Program finished with exit code 0
Press ENTER to exit console.

```